

Clinical Trials Phase Transition

A Survival-Analytic Approach on Public Registry Data

Dr. Valerio Ganci
vganci26@gmail.com

ABSTRACT

Estimating how likely a drug candidate is to advance to its next clinical development phase is a recurring question typically answered through static benchmarks, expert judgment, or costly proprietary platforms. This paper presents a pipeline that derives such estimates directly from public data (National Library of Medicine ClinicalTrials.gov registrations, MeSH ontology, and RxNorm drug vocabulary) resolved into 51,388 drug–disease–phase programs. Because most programs have not yet reached a resolvable outcome, progression is modeled as a right-censored time-to-event process: phase- and disease-specific Kaplan-Meier curves yield a continuous progression probability per program, which is then shrunk toward phase-specific priors via Empirical Bayes to control small-sample noise, producing a posterior rate with a credible interval for every program. Programs are classified into five strategic profiles combining rate, evidence volume, and precision. Results show progression probability declining across phases (21–26% at Phase 1 versus 9–15% at Phase 2/3), heavy oncology activity with favorable early-phase rates, and thinly-evidenced but high-rate outliers, notably a glaucoma program at 20.7% on just four trials, that a rate-only ranking would miss. Findings are directionally consistent with, but more conservative than, published phase-transition literature, a gap attributable to the pipeline’s narrower, registration-based progression definition.

KEYWORDS

Clinical trial progression modeling; Survival analysis; Kaplan–Meier estimator; Empirical Bayes shrinkage; Public data

1 INTRODUCTION

Given a drug candidate currently active in a given clinical phase, for a given disease, how likely is that program to advance to the next phase of development? This is a question that comes up constantly in drug development, and it matters well beyond scientific curiosity: it feeds into decisions about where to focus limited resources, which programs deserve continued investment, and which disease areas are comparatively under-studied relative to how promising they look. Getting a directionally correct, evidence-based answer is worth far more than intuition or anecdote, since a wrong estimate for one program is a minor inconvenience, but the same systematic bias applied across many programs at once is no longer a minor inconvenience at all.

In practice, this question is answered today namely via: published, static probability-of-success tables (e.g. [9]), which are useful benchmarks but are industry-wide averages, not specific to a given disease area, drug class, or time period; internal expert judgment, which is

valuable but slow to scale, hard to audit, and inconsistent across therapeutic areas and reviewers; expensive commercial intelligence platforms (e.g. Citeline/Pharmaprojects, GlobalData), which aggregate this information but are opaque in methodology and costly to license. There is a clear gap for a transparent, reproducible, data-driven estimate of phase-progression likelihood that we could interrogate, recompute, and defend on methodological grounds.

This project builds an end-to-end pipeline (available at [Github link](#)) that ingests public National Library of Medicine (NLM) ClinicalTrials.gov (CTGov) trial registrations and derives, for every observed (drug, disease, phase) program, a statistically grounded estimate of the probability that the program progresses to the next development phase within an empirically calibrated time window, along with a defensible measure of confidence in that estimate. The pipeline does not estimate regulatory approval probability, commercial success, or true "success" in any clinical-outcome sense. This distinction is maintained consistently throughout the project.

Concretely, the pipeline answers three nested questions:

- (1) Program level: For drug X in disease Y at Phase Z, what is our best estimate of its probability of advancing, and how confident are we?
- (2) Disease area level: Which disease areas, across all drugs in our pipeline, show the most favorable (or worst) historical progression dynamics at each phase?
- (3) Portfolio level: which specific programs stand out as high-confidence bets, high-risk/high-reward bets, or emerging signals worth early attention?

The end deliverable is not a single number but a full decision-support layer: a scored, ranked, and profiled table of ~51,000 real-world drug-disease-phase programs, built entirely from public data, with every step of the methodology auditable in code.

2 DATA

The pipeline is built entirely on public, freely accessible data sources. Four inputs are combined: three CTGov tables, the NLM Medical Subject Headings (MeSH) thesaurus, and a subset of the NLM vocabulary of clinical drugs’ normalized names RxNorm. This section describes each source, why it was chosen, and what it contributes to the final analysis.

2.1 CTGov Trial Registrations

ClinicalTrials.gov is the U.S. NLM’s public registry of clinical trials and is the de facto global standard for trial disclosure [1]. It has been mandatory for U.S. interventional trials since 2007 [10] and is used worldwide, making it the largest single source of structured

trial metadata available without a commercial license.

Three CTGov-derived tables are sourced from the Aggregate Analysis of ClinicalTrials.gov database [8]:

- Studies table: one row per registered trial. This table anchors every study in time (start/completion) and in development stage (phase), and provides the status field used to distinguish successful completion from termination or withdrawal.
- Interventions table: one row per intervention within a trial. This is the raw source of drug names as entered by trial sponsors, i.e. free text, inconsistently formatted, and not yet linked to a canonical drug identity.
- Browse conditions: one row per disease/condition term associated with a trial, as tagged by CTGov’s own MeSH-based indexing. This is the raw source of disease identity per trial.

The tables are cached for reproducibility and freely downloadable at the following links: [AACT Studies table](#), [AACT Interventions table](#), [AACT Browse Conditions table](#).

2.2 NLM MeSH Descriptor Ontology

Medical Subject Headings is the NLM controlled hierarchical vocabulary for biomedical concepts [3], and is the same ontology CTGov itself uses to index conditions. The full descriptor set is downloaded and parsed directly.

From each DescriptorRecord, the pipeline extracts:

- The canonical descriptor name and its unique MeSH ID (DescriptorUI).
- Every listed tree number (a MeSH descriptor can appear in more than one place in the hierarchy; the first alphabetically sorted tree number is kept as the canonical placement).
- The MeSH tree group, derived as the top-level segment of the tree number (e.g. tree number "C04.588" → group "C04", the "Neoplasms" branch), which becomes the disease-area aggregation unit used throughout the analysis.
- Every synonym term listed under the descriptor’s concept and term list, so that non-canonical phrasings of a disease name can still be mapped back to the correct descriptor.

This ontology is what allows a free-text CTGov condition tag like "non-small cell lung carcinoma" to be resolved to a stable disease identity (MeSH ID) and a coarser disease-area grouping (MeSH tree group) suitable for portfolio-level aggregation.

The table is cached for reproducibility and freely downloadable at the following link: [NLM MeSH table](#).

2.3 RxNorm Drug Vocabulary

RxNorm is the NLM’s normalized naming system for clinical drugs [7], and is used here to resolve free-text intervention names (e.g. "pembrolizumab", "Keytruda", "MK-3475") down to a single canonical drug identity (RxCUI). Rather than querying the RxNorm REST API live during the main pipeline run, the vocabulary is pre-built once via a separate script ([Github script link](#)) and cached locally

(a copy is available at [Github cached csv file link](#)), for two reasons: firstly, the RxNorm API has per-request rate limits and no bulk-download endpoint for this use case, making a live call per intervention infeasible at 317,000 unique drug-name rows; and secondly, caching guarantees a stable, reproducible vocabulary across pipeline re-runs, rather than one that can silently drift between runs.

The builder script performs a multi-stage enrichment: it (a) resolves an RxCUI for each unique drug name observed in the CTGov Interventions table via the RxNorm exact-match endpoint, (b) fetches the term-type (tty) for each resolved RxCUI and restricts to ingredient-level types (IN, PIN, MIN) so that specific salts or formulations collapse to their active ingredient, and (c) expands each ingredient RxCUI to its full set of known synonyms via the "*all related concepts*" endpoint, so that brand names, generic names, and investigational codes for the same molecule all resolve to one RxCUI. The final cached vocabulary (9,800 entries) contains the following info: drug name, RxCUI, and tty.

3 METHODOLOGY

This section explains, stage by stage, how raw CTGov, NLM MeSH, and RxNorm data is transformed into scored, ranked drug-development programs, and, for every major step, why that specific method was chosen. Given its length, the methodology is split into five subsections:

- 3.1 Raw data preprocessing
- 3.2 Defining "success": the survival master dataset
- 3.3 The Kaplan-Meier progression probability
- 3.4 Empirical Bayes shrinkage
- 3.5 Strategic profiles

3.1 Raw Data Preprocessing

3.1.1 CTGov Studies table: filtering to an analyzable cohort. Before applying any filter, the key text fields are standardized using a common routine. This process converts text to lowercase, removes accents and Unicode variants (NFKD normalization with ASCII folding), and harmonizes British and American spelling according to the spelling map (e.g., haematology → hematology, tumour → tumor) available at the following link: [Github link](#). This normalization ensures that records differing only in capitalization, encoding, or regional spelling are treated consistently during filtering, avoiding the inadvertent exclusion of otherwise valid studies.

After text normalization, four filters are applied:

- (1) Study type filter: only interventional studies are retained. Observational studies are excluded because they do not follow a phase-based clinical development pathway (e.g., Phase 1 → Phase 2 → Phase 3) and therefore cannot contribute to analyses of phase progression.
- (2) Status filter: studies are restricted to the following statuses: completed, terminated, withdrawn, and suspended. This intentionally includes unsuccessful programs, rather than only completed trials. Including both successful and failed studies

is essential for survival analysis, as excluding failures would introduce substantial survivorship bias and lead to overly optimistic estimates of phase progression.

- (3) Phase filter: only studies in Phase 1, Phase 1/2, Phase 2, Phase 2/3, and Phase 3 are retained.
- (4) Temporal filter: studies are limited to those with a start date between 2000 and 2025. Trials initiated before 2000 were excluded because early CTGov records contain substantially less complete information. Studies beginning after 2025 were excluded because they would not have had sufficient follow-up time to reach an observable outcome by the administrative censoring date (2026-06-01), the date on which the underlying CTGov, NLM MeSH, and RxNorm data were downloaded and cached for this analysis, and after which no further trial updates are reflected in the dataset.

3.1.2 Completion evidence. A study with a labelled overall status as "*completed*" is not, by itself, strong evidence that useful results exist. The pipeline computes an "*effective completion date*" ("*primary completion date*", falling back to "*completion date*") and a "*has results after completion*" flag (true when results were posted on or after that effective completion date). A study is flagged "*completed with evidence*" when its overall status is "*completed*" and one of the following conditions is true: it has results posted after completion or it has any effective completion date at all. This flag is what later decides whether a study is allowed to count as "the completed anchor" for a program and, ultimately, whether a program is eligible to register a progression event at all (see Section 3.2).

3.1.3 CTGov Interventions table: resolving drug identity. Before applying any filter, the key text fields are standardized using the same routine used for the CTGov Studies table. After text normalization, the table is filtered and matched to the RxNorm vocabulary:

- (1) Intervention type filter: only interventions classified as drug are retained, removing approximately 59.6% of the original records. Non-pharmacological interventions, including devices, procedures, behavioral interventions, dietary supplements, and other intervention types, are excluded because they do not represent drug development programs.
- (2) Numeric name filter: records with purely numeric intervention names are removed, as these typically correspond to catalogue identifiers or lot numbers rather than valid drug names.
- (3) Comparator filter: intervention names corresponding to comparator arms (namely, placebo, control, sham, observation, or cohort) are excluded. These records describe study controls rather than investigational drug candidates.
- (4) Combination therapy filter: intervention names containing delimiters such as *commas*, math symbols (+), *and*, *with*, *combination*, or *or* are removed. These strings generally represent multi-agent regimens (e.g., Drug A + Drug B) that cannot be reliably decomposed into their constituent drugs without introducing attribution errors. Rather than attempting to infer individual components, the pipeline excludes these ambiguous entries.

- (5) Exact RxNorm matching: the remaining intervention names are matched against the RxNorm vocabulary using exact string matching, successfully resolving approximately 46.5% of drug names to a valid RxNorm Concept Unique Identifier (RxCUI).
- (6) Fuzzy RxNorm matching: drug names that remain unmatched are subsequently compared against the same vocabulary using RapidFuzz's token set ratio algorithm with a similarity threshold of 90 (RapidFuzz). This second pass increases the overall matching rate to approximately 64.5%. Fuzzy matching is included to account for common inconsistencies in sponsor-entered intervention names, such as typographical errors, additional dosage information (e.g., pembrolizumab 200 mg), or minor formatting differences. A deliberately conservative similarity threshold minimizes the risk of incorrectly assigning different drugs to the same RxCUI.
- (7) Final unresolved-name filter: any intervention that cannot be assigned a valid RxCUI after both the exact and fuzzy matching stages is excluded from subsequent analyses. An unresolved drug identity cannot be aggregated into a program (which are defined by RxCUI), and a synthetic "unknown" bucket would just merge many unrelated candidates together, corrupting every downstream rate estimate for that bucket. Exclusion is the more statistically honest choice, at the cost of coverage.

3.1.4 CTGov Browse Conditions table: resolving disease identity. The key text fields are standardized using the same routine used for the CTGov Studies table. The CTGov Browse Conditions table is then merged against the NLM MeSH ontology on exact normalized term match. Two additional decisions matter here:

- (1) Mapping priority: when the same MeSH term could match either a MeSH descriptor's own name or one of its synonyms, the descriptor match is always preferred. A descriptor name is the unambiguous canonical label for a concept; resolving ties toward it avoids arbitrary tie-breaking when a term happens to be a synonym of more than one descriptor.
- (2) Deduplication of MeSH-list vs. MeSH-ancestor: CTGov tags each condition as either "MeSH-list" (a term the sponsor directly selected) or "MeSH-ancestor" (a broader term CTGov's indexer inferred automatically as a parent of the selected term). For a given study, if any MeSH-list info exist, only those are kept; only when a study has no MeSH-list info at all does the pipeline fall back to its single deepest (most specific) MeSH-ancestor. MeSH-list terms are sponsor-intended and specific; keeping both MeSH-list and MeSH-ancestor info for the same study would duplicate that study's disease signal (once at the specific level, again at every ancestor level up the tree), artificially inflating counts for broad disease groups. Falling back to the single deepest ancestor when no MeSH-list exists preserves coverage for studies that were only auto-indexed, while keeping specificity as high as the data allows.

This step achieves 99.7 % study coverage and 77.0 % unique-term coverage, i.e. almost every trial retains a resolvable disease identity

even though roughly a quarter of the long tail of rare condition terms do not map.

3.1.5 Net effect of raw data preprocessing. Starting from 583,450 raw studies, the full filter cascade (study type, status, phase, temporal) plus the completion/evidence computation retains 120,674 interventional drug studies spanning Phase 1-3, 2000-2025. These, joined against resolved drug and disease identities, are what feed the later steps of program-level aggregation and survival dataset construction.

3.2 Defining "Success": the Survival Master Dataset

The most naive way to answer "did this program progress?" would be to pick a fixed time window (e.g. "did a next-phase trial start within 3 years?") and assign every program a hard 0 or 1. This has two serious problems:

- (1) Right-censoring: a large share of programs are still ongoing, or ended too recently, for their true outcome to be knowable yet. Labeling these as a hard "0" (no progression) would systematically undercount progression, penalizing recently-started programs simply for not having had enough time to progress.
- (2) An arbitrary window choice would impose a single threshold that is only nominally defensible on every downstream number, and different phases plausibly need different windows (Phase 1 → Phase 2 transitions tend to happen faster than Phase 2 → Phase 3).

Instead, this work adopts a survival (time-to-event) analysis framework, the standard statistical approach for data in which outcomes may either be observed (the event occurs at a known time) or right-censored (the event has not occurred by the end of follow-up time but may still occur in the future). Under this framework, every program contributes information according to its observed follow-up time, rather than being forced into a binary outcome based on an arbitrary time threshold. As a result, both successful programs and those still under observation are incorporated appropriately, avoiding the bias introduced by fixed follow-up windows. This handling of censoring is a key methodological strength of the pipeline and provides the foundation for estimating progression probabilities using the Kaplan–Meier estimator (Section 3.3).

3.2.1 Program-level aggregation. In this work a program is defined as RxCUI - MeSH ID - Phase triple, i.e. one specific drug, in one specific disease, at one specific development phase. All studies sharing that triple are aggregated into one program, with the following info:

- the number of distinct trials that support this program
- the number of distinct supporting trials that were "*completed with evidence*" as defined in Sect. 3.1.2
- the number of distinct supporting trials that failed outright
- the earliest start date across all studies in the program
- the latest "*effective completion date*" across all studies as defined in Sect. 3.1.2

- the earliest completion date among only the "*completed with evidence*" studies

A single drug–disease combination may be represented by multiple clinical trials within the same phase, for example across different patient populations, geographic regions, or study designs. Treating these trials as independent observations would overrepresent programs that have been evaluated in multiple studies. Aggregating trials into a single program-level entity ensures that each drug–disease pair is represented once, allowing the analysis to estimate progression at the level of the development program rather than the individual study. This provides a more appropriate unit of analysis for modelling phase escalation across the clinical development pipeline.

This choice is also, in an important sense, forced by the nature of the public data being used, not just a modelling preference. CTGov records which studies exist and when they started and ended, but it does not record which specific completed study caused a sponsor's decision to advance a drug-disease pair to the next phase. Given that, this pipeline cannot claim, and does not attempt to claim, that any single study "progressed" to the next phase in a causal sense. What can be observed at the program level is a weaker but still useful signal: that a given drug-disease pair had at least one completed study in the current phase, and that a study for the same pair in the next phase started within a plausible time window afterward (see Section 3.2.2). Aggregating to the program level is therefore the most honest unit given what the underlying data can actually support; treating individual studies as the unit of analysis would imply a causal precision about which study led to which outcome that this data source simply cannot provide.

3.2.2 The empirical max gap: how long is "long enough" to wait. To decide whether a next-phase study "counts" as a progression of a given program, the pipeline needs a maximum allowable time gap between the current-phase program's start and the next-phase program's start. Rather than imposing a single literature-default gap, the pipeline computes this empirically, per phase, from the data itself in the following way: for every program with at least one completed study, and for every later-phase program for the same (RxCUI, MeSH ID) pair that starts on or after it, the observed gap is recorded. The per-phase 90th percentile of these observed gaps becomes that phase's max gap.

Using the 90th percentile avoids letting a handful of extreme outlier gaps expand the window arbitrarily, while still covering the large majority of genuine escalations. Different phases empirically show different typical gaps (e.g. Phase 2 shows a longer 90th percentile than Phase 1 in the observed data), so a single global constant would either be too short for later phases (misencoding real progressions as failures) or too long for earlier phases (letting unrelated, much-later programs count as "progression"). When a phase has fewer than 10 observed gaps the pipeline falls back to a fixed default of 3,650 days (10 years), a figure consistent with published estimates of total time spent in clinical development, which range from roughly 6 years for non-oncology programs to over 13 years for oncology programs [9].

3.2.3 Building the survival dataset: duration and event. A survival dataset is therefore constructed in which each row represents a unique drug–disease–phase program, defined by a combination of RxCUI, MeSH ID, and clinical phase. Programs are created only for Phase 1, Phase 1/2, Phase 2, and Phase 2/3, as these phases have a subsequent phase to which they can progress. Phase 3 programs are not included as analysis units because they represent the final stage of the progression pathway and therefore have no downstream phase within the framework. This is a deliberate scope boundary rather than an oversight; Phase 4 studies (post-marketing surveillance) presuppose that regulatory approval has already occurred, an event this pipeline has no visibility into and that depends on a regulatory decision CTGov does not record.

Each program is assigned a follow-up time and an event indicator according to the observed clinical development history:

- Confirmed progression (event = 1): a program is considered to have progressed if it contains at least one completed study with evidence of completion (see Section 3.1.2) and a corresponding program for the same drug–disease pair is subsequently observed in the next clinical phase within the maximum allowable phase-transition interval. The follow-up time is defined as the number of days between the start of the current phase and the start of the subsequent phase.
- Status-confirmed non-progression (event = 0): programs for which all studies are either terminated or withdrawn are treated as censored observations. Follow-up ends at the latest available completion date, or at the administrative censoring date (2026-06-01) if no completion date is available. Although these programs did not progress, they are not treated as a separate failure event. The analysis is specifically designed to model time to phase progression, and therefore only confirmed progression constitutes an event. All other outcomes are handled through right censoring, allowing the analysis to remain within the standard single-event Kaplan–Meier framework rather than requiring a competing-risks model.
- Ongoing or unresolved programs (event = 0): all remaining programs are also treated as right-censored. Their follow-up extends to the administrative censoring date (2026-06-01), reflecting that no progression had been observed by the end of the observation period, while allowing for the possibility that progression may still occur in the future.
- Excluded programs: programs with non-positive follow-up times are removed, as they do not provide meaningful information for time-to-event analysis.

3.2.4 Net effect of the survival analysis framework. The result (51,388 programs, 8,599 events, 16.7% event rate) is a threshold-free, right-censoring-aware dataset in which:

- a program that just started and hasn’t progressed yet is not penalized as a “failure” and instead it correctly contributes only a short, partially-informative follow-up period.
- a program that failed outright is censored at a meaningful, data-derived date rather than an arbitrary one.

- the escalation window itself is grounded in the same data being modelled, rather than an external assumption

The observed 16.7% event rate is consistent with published Phase 1 → 2 probability-of-success benchmarks in the literature (roughly 10–25% [9]), which is a useful sanity check that the pipeline’s definitions are producing plausible numbers.

3.3 The Kaplan–Meier Progression Probability

The survival dataset described in Section 3.2 assigns each program a follow-up duration and an event indicator, but summarizing these data with a single overall progression rate (16.7%) would ignore the temporal dimension of the problem. Such an approach would treat a program that failed to progress after 100 days identically to one that remained under observation for 3,000 days without progressing, despite the latter providing substantially more information about its progression potential. To account for both the timing of events and right-censored observations, progression probabilities are estimated using the Kaplan–Meier (KM) estimator [6]. In this context, the survival function, $S(t)$, represents the probability that a program has not yet progressed beyond its current phase by time t . Because the KM estimator makes no assumptions about the underlying hazard function, it avoids imposing an arbitrary parametric distribution on phase progression. Given the absence of strong biological or empirical evidence supporting a specific hazard shape, this non-parametric approach provides a robust and well-justified basis for estimating progression probabilities.

3.3.1 Phase level Kaplan–Meier curves. Separate KM survival curves are estimated for Phase 1, Phase 1/2, Phase 2, and Phase 2/3, using each program’s follow-up duration and event indicator within the corresponding phase. For each phase, the analysis estimates the complete survival function, $S(t)$, together with summary statistics used for model diagnostics, including the median time to progression and the estimated probability of remaining in the current phase after 5 and 10 years.

Phase-specific models are used because progression dynamics differ substantially across the clinical development pipeline. Transitions from Phase 1 to Phase 2, for example, involve different trial durations and regulatory requirements than transitions from Phase 2 to Phase 3. Estimating a single survival curve across all phases would therefore obscure these differences and produce progression estimates that are not representative of any individual phase.

Across all four phases, the estimated median time to progression was not reached, indicating that fewer than half of the observed programs had progressed during the available follow-up period. This finding is consistent with the relatively low overall progression event rate (approximately 16.7%). Estimated 5-year survival probabilities ranged from 78.5% to 91.1%, equivalent to a 5-year cumulative progression probability of roughly 8.8% to 21.5% depending on phase. These figures are directionally consistent with published phase-transition literature [5, 9] in that progression probability declines from earlier to later phases, but they sit meaningfully below the absolute transition rates reported in that literature; the gap is discussed in detail in Section 5.1.

3.3.2 Phase - disease-area level Kaplan-Meier curves. KM curves are estimated also for each combination of clinical phase and MeSH disease tree group (e.g., Phase 2 – Neoplasms or Phase 2 – Nervous System Diseases). These curves are fitted only for groups with at least 10 observed progression events; groups with fewer events are not modelled.

A minimum-event threshold is used because KM estimates become highly unstable when based on very small numbers of events. In sparse groups, the occurrence or removal of a single progression can substantially alter the estimated survival curve, producing unreliable progression estimates. Requiring at least 10 observed events provides a conservative safeguard against overfitting and ensures that disease-specific survival curves are estimated only where sufficient empirical evidence is available. In the present dataset, 69 phase-disease groups met this criterion, while 32 were considered too sparse for independent modelling. For these groups the analysis defaults to the corresponding phase-specific survival curve, providing a more stable estimate than fitting an unreliable disease-specific model or excluding the program from analysis altogether.

Finally, disease-specific survival models are included because progression dynamics are not determined by clinical phase alone. Even within the same phase, development timelines and probabilities of advancement vary considerably across therapeutic areas due to differences in disease biology, trial design, endpoint selection, and regulatory requirements. Estimating separate curves for well-supported disease groups allows these differences to be captured where the data justify it.

3.3.3 Computing Kaplan-Meier progression probability. Each program is assigned an individual KM progression probability by evaluating the most appropriate survival curve at its observed follow-up time. Whenever available, the disease-specific curve corresponding to the program’s clinical phase and MeSH disease tree group is used; otherwise, the analysis falls back to the phase-specific survival curve. The progression probability is then calculated as:

$$P(\text{progression}) = 1 - S(t),$$

where $S(t)$ is the probability of remaining in the current phase after an observed follow-up time t . Evaluating the survival curve at each program’s own follow-up duration transforms the population-level survival model into an individualized, continuous estimate of progression probability. Programs that have remained in the same phase for an extended period without progressing receive higher progression probabilities than recently initiated programs, reflecting the increasing evidence against eventual advancement as follow-up accumulates. Consequently, this measure is not a binary success/failure label but a continuous value between 0 and 1 that incorporates both observed progression events and the information contained in right-censored observations. These individualized progression probabilities serve as soft labels in the subsequent Empirical Bayes shrinkage analysis (Section 3.4), allowing downstream estimates to account for both observed outcomes and the varying amounts of follow-up information available for each program.

3.4 Empirical Bayes Shrinkage

Imagine two drug-disease pairs. Pair A has been studied in just one trial, and that trial happened to show progression, so its raw rate looks like a perfect 100%. Pair B has been studied in forty trials, of which nine showed progression, a raw rate of 22.5%. If programs were simply ranked by their raw KM progression probability, pair A would shoot to the very top of the list, ahead of pair B, purely because it got lucky with a tiny sample. That is obviously the wrong conclusion: pair A’s “100%” tells us almost nothing reliable, while pair B’s 22.5% is backed by real, substantial evidence. For this reason, we apply an Empirical Bayes (EB) shrinkage.

The intuition behind EB shrinkage is simple, even if the formulas look technical: instead of trusting each group’s own raw rate in isolation, pull it partway toward a sensible “average” rate that reflects everyone else with similar characteristics, and pull it harder the less evidence that group has of its own. A pair with only one study gets pulled almost all the way to the average, because one data point is barely more informative than no data point at all. A pair with forty studies barely moves, because it already has enough evidence to speak for itself.

Returning to the example above: instead of a raw 100%, pair A’s rate would be shrunk down toward the broader average, landing somewhere in the range the data actually supports, once its lone lucky trial is placed in proper context. Pair B’s 22.5%, already well-supported, would barely change. The net effect is a much fairer ranking: programs are separated by genuine signal, not by which ones happened to get a small, noisy sample.

Under the hood, this is implemented using a standard statistical recipe, a Beta-Binomial model, which is simply the natural mathematical tool for “success rates between 0% and 100%, with the ability to add real evidence into a starting assumption” [4]. It has the convenient property that combining a starting assumption (“the prior”) with observed evidence always produces a new, updated rate (“the posterior”) through simple arithmetic, with no heavy computation required.

Before any individual pair can be shrunk, the pipeline first needs to decide what “the average” is, the anchor point that under-evidenced pairs get pulled toward. This anchor, called the prior, is estimated separately for each phase. Two things go into building each phase’s prior: (a) a central value, the overall average KM progression probability observed across that phase, which becomes the rate that under-evidenced pairs get pulled toward; (b) a sense of how much genuine disease-to-disease variation exists within the phase, versus how much of the spread in the raw data is just noise. This is estimated from how much rates vary across different disease groups (MeSH ID), but only using disease groups that have more than one program to compare, since a disease group represented by a single program has no internal variation to measure at all, and including it would mistake one lucky or unlucky data point for genuine disease-level variation, throwing off the whole estimate.

The less genuine variation there turns out to be across disease groups within a phase, the more the pipeline trusts the phase-wide average and the harder it pulls under-evidenced pairs toward it.

This logic, left unchecked, has a failure mode: if a phase happens to show very little observed variation across its disease groups, whether because the variation is genuinely small, or simply because too few multi-program disease groups exist to measure it reliably, the estimated prior strength could grow arbitrarily large, to the point where the phase-wide average would dominate almost every pair’s posterior, no matter how much evidence that specific pair has of its own. To guard against this, the pipeline caps the prior’s strength at a fixed ceiling of 100. This specific value was chosen to sit comfortably below the pipeline’s own upper range of real evidence (individual pairs in this dataset range up to roughly 190 supporting trials) so that even a prior estimated as extremely confident can still, in principle, be overridden by a well-studied pair, rather than permanently overruling it.

3.4.1 Shrinking individual drug-disease pairs. With a per-phase anchor in place, every individual drug-disease pair is shrunk toward its own phase’s anchor, weighted by how many actual clinical trials support it. The result for each pair is a shrunk rate (EB rate) together with a 95% confidence range around it. That range matters just as much as the rate itself: two pairs can share the same EB rate while one is backed by dozens of trials (a narrow, confident range) and the other by just a handful (a wide, uncertain range).

3.4.2 Shrinking whole disease areas. The identical shrinkage logic is applied also at a coarser level: instead of individual drug-disease pairs, whole disease areas (e.g. all oncology programs within a phase) are shrunk toward the same kind of phase-level anchor. The only real difference is what counts as "evidence" at this level; here, it is simply how many distinct programs exist within that disease area, rather than how many clinical trials back a drug. This avoids letting a handful of unusually well-studied drugs make an entire disease area look more evidence-rich than it really is.

3.5 Strategic Profiles

At this point every drug-disease pair and every disease area has a shrunk, evidence-aware progression rate and a sense of how confident that rate is. The final step is to turn this into a metric by which we can actually rank programs and disease areas.

The pipeline keeps this step deliberately simple: the ranking score used for every unit is exactly its EB rate, nothing more. The reasoning is straightforward. Think of a simple go/no-go investment decision, where the only question is "will this program advance or not." The expected value of making that bet is just the probability that it pays off, so, absent any outside information about how much a given program would actually be worth if it does advance (peak sales potential, strategic fit, cost to reach the next milestone, and so on, none of which this pipeline has access to), the shrunk progression probability itself already is the right thing to rank on.

An alternative approach would have been to build one composite score that blends the rate together with how much evidence backs it and how confident the estimate is, for example, rewarding a well-evidenced 20% rate over a thinly-evidenced 25% rate by some formula. The pipeline deliberately avoids this: folding confidence and evidence volume into the ranking number itself would require

picking arbitrary weights for how much each should matter, and would hide the underlying trade-off rather than showing it. Instead, evidence volume (N_{trials}) and estimate precision (CI_{width}) are kept fully visible and untouched, and are used directly to build the richer classification described next.

Rather than reducing every program to a single rank, each program is classified into a *strategic profile*, a richer label built from three binary signals, each itself computed per-phase:

- Sufficient evidence: $N_{trials} > 3$
- High EB rate: EB rate at or above its phase’s 75th percentile
- Low uncertainty: CI_{width} at or below its phase’s 25th percentile

These combine into five mutually exclusive profiles:

Table 1: Strategic profile definitions

Profile	N_{trials}	EB rate	CI_{width}
High confidence	> 3	$\geq 75th\ Q$	$\leq 25th\ Q$
High risk high reward	> 3	$\geq 75th\ Q$	$> 25th\ Q$
Emerging signal	≤ 3	$\geq 75th\ Q$	X
Stable but modest	> 3	$< 75th\ Q$	$\leq 25th\ Q$
Standard	*	*	*

Notes. Q: Percentile; X no condition required; * remaining combinations.

This classification is what gives each program a qualitative read on how it is a good or bad prospect, not just whether it is one: two programs with an identical EB rate can carry very different profiles depending on how much evidence and how much certainty stand behind that shared number, and it is this profile, not the raw rate alone, that is used to organize and interpret the results in Section 5.3.

4 LIMITATIONS

A pipeline that only shows its strengths is less credible than one that also states clearly where it approximates, where it assumes, and where it is simply out of scope. In this Section, the limitations of the work are presented, grouped into three categories: input data, methodology, and the results themselves.

4.1 Input data limitations

4.1.1 CTGov is not a complete registry of global drug development. CTGov registration has been mandatory for most U.S. interventional trials only since 2007. Trials that ran entirely outside the U.S., under non-IND regulatory pathways, or before the mandate took effect, may be absent or incompletely recorded. Additionally, a drug-disease program can genuinely progress, through licensing, acquisition, or a sponsor’s internal decision to advance, without that progression ever surfacing as a new CTGov trial registration in a detectable time window (e.g. an EMA-only European trial, or a deal that delays the next study’s registration). Any such progression is invisible to this pipeline and will be counted as censored, not as an event.

4.1.2 Completion label bias. Whether a completed study has results posted, plausibly correlates with the study's outcome; positive, newsworthy results may be reported faster or more consistently than negative ones. Since our label "*completed with evidence*" partly depends on this flag, this could introduce a bias toward preferentially recognizing successful-looking studies as "complete" ahead of unsuccessful ones, though the practical impact is limited by the fact that the flag also accepts studies with just an effective completion date. On that regard, as noted in Section 3.1.2, "*completed with evidence*" is true when a completed study either has post-completion results or simply has any effective completion date. This means a study with a completion date but no posted results still passes as "evidence of completion," which is a looser bar than the flag's name suggests.

4.1.3 RxNorm coverage gap. Roughly a third of intervention-name instances never resolve to an RxCUI, even after exact and fuzzy matching, and are excluded entirely rather than retained under a synthetic "unknown drug" identity. These unresolved names are likely disproportionately novel/investigational compounds, biosimilars, and company-internal code names not yet indexed in RxNorm, meaning the analyzed dataset likely skews toward more established, already-named compounds and may under-represent frontier/early-stage chemical matter.

4.1.4 MeSH tree group is a coarse disease unit. MeSH tree group is the top-level segment of a MeSH tree number (e.g. "C04" covers all of Neoplasms). This means clinically very different diseases (for instance, breast cancer and leukemia) can share the same tree group. The pair-level analysis uses the finer MeSH ID (specific descriptor), but the tree-level curves, and EB priors all operate at this coarser group level, trading disease specificity for having enough events per group to fit a stable KM curve.

4.2 Methodology limitations

4.2.1 "Progression" is defined narrowly. A progression event requires a next-phase study for the same (RxCUI, MeSH ID) pair to start within the empirically-derived max gap window. This single definition misses several real-world progression patterns: licensing-out to another sponsor who registers under different identifiers, or progression that is confirmed by non-trial evidence (e.g. a regulatory filing) rather than a new trial registration. All of these would be miscoded as censored (non-events) under the current definition, not as progressions.

4.2.2 The empirical max gap window is conservative by construction. Using the 90th percentile of observed escalation gaps (rather than, say, the 99th percentile or the observed maximum) means roughly the slowest 10 % of true progressions, by design, fall outside the detection window and are miscoded as censored. This is a known, intentional trade-off (favoring stability of the percentile estimate over maximal recall), but it does mean the pipeline's event rates and progression probabilities are systematically biased slightly downward, not unbiased in either direction.

4.2.3 No competing-risks modelling. The survival framework used here treats every non-progression outcome (status-confirmed failure and ongoing/unknown alike) as simple right-censoring within

a single-event model. A more granular alternative would model "confirmed termination/withdrawal" as a distinct competing risk rather than folding it into the same censoring mechanism as "still ongoing", since these are conceptually different situations (one is a resolved negative outcome, the other is simply unresolved). A competing-risks model (e.g. cumulative incidence functions) could in principle separate them. The current single-event KM approach was chosen for manageability and interpretability, at the cost of this distinction.

4.2.4 Drug identity conflation risk. RxNorm resolution collapses different trade names, generic names, and investigational codes for what should be the same active ingredient into one RxCUI, restricted to ingredient-level term types (IN/PIN/MIN) in the vocabulary build. This is generally the correct behavior, but two opposite risks remain: (a) different formulations or delivery mechanisms of what is nominally "the same" molecule (e.g. an oral vs. an injectable form) may be merged into one program even though their clinical development paths can meaningfully differ, and (b) the fuzzy-matching step (threshold = 90) could, in rare cases, still merge two genuinely different but textually similar drug names. The exact-match-first, conservative-threshold design (Section 3.1.3) minimizes but does not eliminate either risk.

4.2.5 Empirical Bayes priors rely on their own moment-based approximation. The phase-level Beta priors are estimated via method-of-moments on group-level rates, not via a fully Bayesian hierarchical fit (e.g. MCMC or a proper hyperprior). Method-of-moments is fast, closed-form, and adequate at this data scale, but it is itself an approximation, particularly sensitive when between-group variance is small or estimated from relatively few multi-program MeSH ID groups, which is part of why a prior strength cap of 100 was necessary as a safety bound (Section 3.4) rather than trusting the raw moment estimate unconditionally.

4.2.6 Per-phase thresholds are data-driven, not fixed. The strategic profile high rate/low uncertainty thresholds are computed as empirical quantiles of the current run's own data (75th/25th percentiles). This is a deliberate design choice (Section 3.5) for within-run comparability, but it also means the meaning of "high" or "low" is relative to the current dataset, not to any fixed, externally-anchored bar: an "high" EB rate program in one pipeline run is not necessarily comparable in absolute terms to an "high" tier program from a future run on an updated dataset, only within that run's own distribution.

4.3 Results limitation

4.3.1 What the results are, restated precisely. EB rate is an Empirical-Bayes-shrunk posterior mean of a survival-derived, continuous progression signal, itself defined relative to a specific, narrow operational definition of "progression" (Section 3.2, 4.2.1) and a specific, conservative detection window (Section 3.2.2, 4.2.2). It is not:

- a historical, empirically-observed raw success rate
- a probability of eventual regulatory approval
- a probability of commercial success

- a causal or risk-adjusted estimate accounting for drug mechanism, sponsor quality, indication difficulty, or trial design quality.

Any consumer of these outputs should treat EB rate as "this pipeline's best data-driven estimate of the likelihood of a next-CTGov-registered-phase-start, given this specific methodology," not as an external, validated ground truth.

It is also worth restating a point already made about how programs are constructed: EB rate is not, and cannot be, a causal claim about any single clinical trial. A public registry like CTGov shows that a study happened and when, but never why a sponsor chose to start the next one; that reasoning lives inside the sponsor's own internal decision process, which no public data source discloses. As a result, EB rate should be read as a probability attached to a drug-disease program as a whole, not as a score for any individual trial within it. Two different Phase 1 studies for the same drug-disease pair, run in different populations or geographies, do not each get their own "progression probability" in this pipeline's output, nor could they responsibly be given one from CTGov data alone; only the program they jointly belong to does.

4.3.2 No uncertainty propagation across pipeline stages. Confidence intervals are computed within the EB shrinkage step from the Beta posterior but do not propagate the earlier uncertainty inherent in the KM curve estimates themselves (e.g. curve-level standard error at a given duration) into the final credible interval. The reported CI_{width} therefore underestimates the pipeline's true end-to-end uncertainty to some degree, since it treats each program's KM progression probability as a fixed input to the EB step rather than itself an estimate with its own error bar.

5 RESULTS

This section presents the results of the pipeline, summarized in three different outputs designed to answer the three nested questions framed in Section 1, at increasing levels of granularity:

- (1) Phase Progression Curves: how does cumulative progression probability evolve over time, separately for each development phase? This serves as the baseline against which any individual disease area or program should be evaluated.
- (2) Disease Area Opportunity Map: across all top-level MeSH disease groups represented in the data, which combinations of phase and disease area show the most (or least) favorable EB-shrunk progression rates, and how much evidence backs each one?
- (3) Top Programs Ranking: among individual drug-disease programs with adequate evidence, which specific candidates rank highest by EB-shrunk progression probability, and what strategic profile does each carry?

5.1 Phase Progression Curves

The cumulative progression probability for each of the four target phases are shown in Fig. 1 as a continuous curve over time since phase start with a shaded 95% confidence band around each curve. At the 5-year mark, cumulative progression probability is 21.5% for Phase 1, 19.8% for Phase 1/2, 10.4% for Phase 2, and 8.8% for

Phase 2/3. By the 10-year mark, these rise to 26.3%, 25.1%, 15.0%, and 11.6% respectively. Two patterns stand out:

- A clear phase ordering: earlier phases (Phase 1, Phase 1/2) show consistently higher cumulative progression probability than later phases (Phase 2, Phase 2/3) at every time horizon plotted.
- A long, slow-rising tail: none of the four curves come close to reaching 50% cumulative progression even at the 11-year horizon plotted, consistent with the "median not reached" diagnostic already noted in Section 3.3.1; most programs in this dataset have not (yet, or ever) progressed within the observed follow-up window.

The declining progression rate from earlier to later phases is directly consistent with the phase-level EB priors already reported in Section 3.4 (Phase 1 mean 19.6%, Phase 2/3 mean 9.7%); the KM curves are, in effect, the more granular time-resolved version of that same signal, and the two are internally consistent, which is a useful cross-check that the pipeline's stages agree with each other. This pattern is also directionally consistent with the general industry understanding that attrition compounds as a drug candidate advances (each additional phase represents a harder bar to clear, on a smaller and more selected population of surviving candidates [2]), even though, as stated in Section 4.3.1, this pipeline is not a formal probability-of-success benchmark study.

Published phase-transition literature reports considerably higher transition rates than this pipeline's curves show at face value. A widely cited reanalysis of trial outcomes from 2006-2015 reports a 63.2% success rate for Phase 1-to-Phase-2 transitions and a 30.7% success rate for Phase 2-to-Phase-3 transitions across all indications combined [5, 9]. This pipeline's own 5-10 year cumulative progression estimates for Phase 1 (21.5%-26.3%) and Phase 2 (10.4%-15.0%) are roughly a third of those published figures.

This gap is expected, not a red flag, once the difference in definitions is made explicit: published phase-transition studies typically define "success" as the phase concluding with a formal sponsor decision to advance, often drawing on curated, proprietary, or company-disclosed program-tracking data with more complete visibility into a program's true fate [5, 9]. This pipeline instead detects progression only when a new CTGov trial registration for the same drug-disease pair appears within an empirically bounded time window (Sections 3.2, 4.2.1, 4.2.2), a strictly narrower and more conservative signal. A program that advances internally, is licensed out, or is confirmed to progress through a channel other than a fresh CTGov registration for the identical (RxCUI, MeSH ID) pair will show up here as censored, not as an event.

5.2 Disease Area Opportunity Map

Picture a map with development phase running left to right, and every major disease area stacked top to bottom, ordered so that the diseases with the strongest overall progression tendency sit near the top and the weaker ones settle toward the bottom. Scattered across this grid are bubbles, one for every disease area at every phase where there is data to show. Each bubble tells two stories at once: its color says how likely programs in that disease area are to

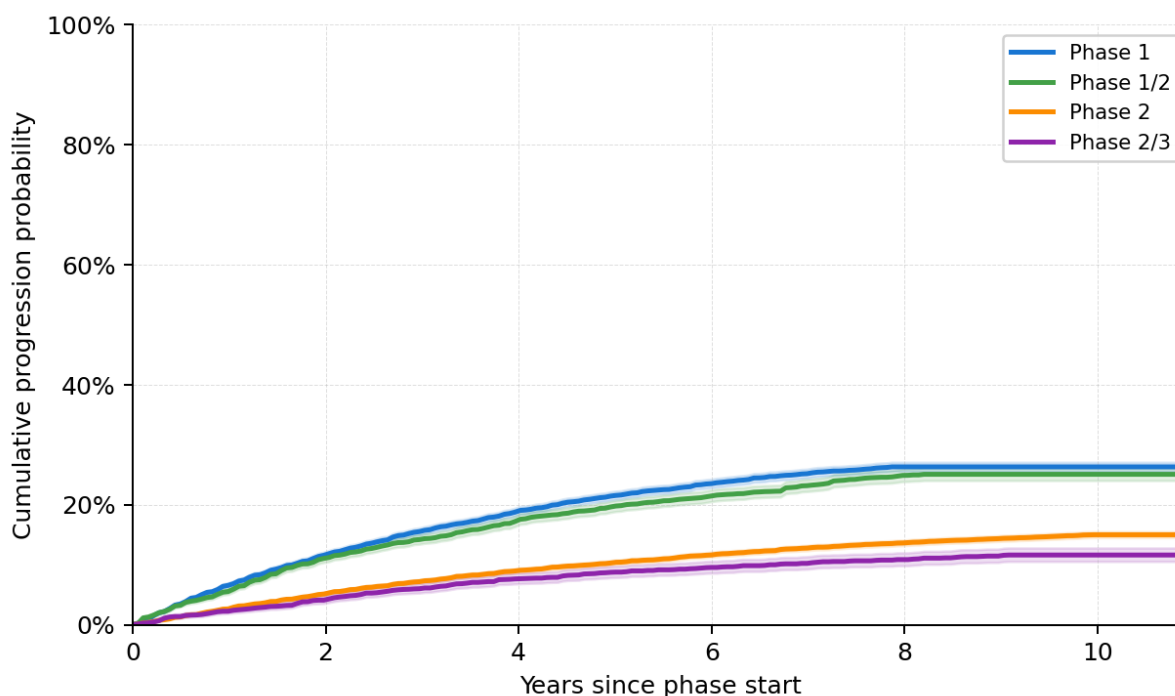


Figure 1: Cumulative KM progression probability as a function of time for each of the four target phases with a shaded 95% confidence band.

progress, and its size says how much real evidence stands behind that number. A big, deeply colored bubble is a well-established, reliable signal; a small, brightly colored bubble is a number worth noticing but not yet worth trusting on its own. This is what is shown in Fig. 2.

One shape dominates the entire map: oncology. Across nearly every phase, the Neoplasms disease area outshines every other one: 8,320 Phase 1 programs, 4,628 in Phase 1/2, and 10,036 in Phase 2, an order of magnitude beyond almost anything else on the chart. This is not a surprise; cancer has long been the single largest area of clinical development activity worldwide, and the map simply makes that dominance visible at a glance. Its progression rates (22.6% at Phase 1, 20.9% at Phase 1/2, 11.4% at Phase 2) sit close to the overall phase averages introduced in Section 5.1, which makes sense by realizing just how much of each phase's total evidence oncology alone is supplying.

But the most interesting bubble on the map is not the biggest one. In the Phase 1 column, a small bubble for Eye Diseases actually outshines Neoplasms in color, showing the single highest rate anywhere on the chart at 25.5%, edging out oncology's own 22.6% in the same column. On its face, that reads as an exciting signal. But its size tells a different part of the story: where Neoplasms rests on 8,320 trials of support, Eye Diseases rests on just 126. This is exactly the tension the map is built to expose: a bright color without the size to back it up is a lead, not a conclusion, and reading one without the other is how a promising-looking number turns into a false lead.

That, in a sentence, is what this map is for: providing an at-a-glance view of where the most promising unexplored opportunities may lie; large, richly colored areas like early-phase oncology as dependable ground, and small, brightly colored areas like Phase 1 Eye Diseases as places worth a second look before anyone commits real capital. In practice, that means interpreting a disease area's size as a measure of the statistical reliability of its estimate: Neoplasms, because it contains the largest dataset, offers the most statistically dependable result on the entire chart, a kind of anchor point against which smaller, noisier bubbles elsewhere can be judged. A small, high-color bubble like Eye Diseases at Phase 1 is best treated as an invite to dig one level deeper, pulling the individual drug-disease programs behind that bubble and checking how well-evidenced each one individually is, rather than as a green light on its own, since a disease area's aggregate rate can just as easily be carried by one or two standout programs as by a genuinely broad trend. And because the map lays every phase out side by side, we can also simply follow one disease area's row from left to right and watch whether an encouraging early rate holds up, improves, or fades as that disease area moves into later phases, a trajectory that no single phase's number, taken alone, could ever reveal.

It is also worth pausing on what oncology's dominance here does, and does not, say about how "easy" cancer drug development really is. Published estimates put oncology's unconditional, end-to-end approval rate at somewhere around 3-5%, among the lowest of any major therapeutic category, with typical development timelines running near 13 years, roughly twice as long as the average for non-oncology programs [9]. Read alongside that broader picture,

this map's story about oncology starts to make more sense: cancer candidates clear the earliest, safety-focused phase relatively often, sometimes because meaningful efficacy signals can already be glimpsed in cancer patients even at this early stage, but the real attrition sets in later, in the harder efficacy and comparative-effectiveness hurdles of Phase 2 and beyond, exactly where this map already shows oncology's own rate beginning to soften, down to 11.4% by Phase 2. This pipeline cannot see all the way through to final approval, bounded as it is by its narrower CTGov-registration-based definition of progression, so its findings here should be read as consistent with, rather than a full replication of, that deeper and longer-running body of evidence.

5.3 Top Programs

Rather than one long list ranked purely by rate, this view is organized as a table (Table 2) grouped by strategic profile (Section 3.5.1): for each of the five profiles ("*high confidence*", "*high risk high reward*", "*emerging signal*", "*stable but modest*", and "*standard*") it shows the ten programs with the highest EB rate carrying that label. The point of organizing it this way is that a program's risk category matters just as much as its raw rate: a 20% rate resting on three trials and a 20% rate resting on seventy are very different propositions, even though a rate-only ranking would place them side by side. Working through the table profile by profile tells a fairly coherent story about where this dataset's confidence actually lives.

The "*high confidence*" group are the safe, proven core of the dataset: gemcitabine in pancreatic cancer leads at 27.1% (76 supporting trials), followed by pemetrexed in non-small-cell lung cancer (25.4%, 43 trials), dexamethasone and lenalidomide both in multiple myeloma (24.9% and 24.8%, on 72 and 43 trials respectively), and a further six programs, cyclophosphamide in melanoma and in chronic lymphocytic leukemia, trastuzumab in breast cancer, erlotinib and fluorouracil in NSCLC and rectal cancer, all clustered in the 24-25% range on 27 to 32 trials apiece. Nothing here is a surprise: every name is an established, extensively studied oncology therapeutic, and the whole group reads as exactly what a "safe, well-evidenced bet" should look like.

The "*emerging signal*" group tell a very different story, and they are arguably the most interesting in the table. Every program here rests on exactly three supporting trials, the smallest amount of evidence the pipeline will still count, yet several show rates in the same 20-21% range as some "*high confidence*" programs: pazopanib in macular degeneration and vitamin E in macular edema both lead at 20.8%, followed closely by levofloxacin in cataract and a run of oncology pairings (oxaliplatin in two different lymphoma subtypes and in hematologic neoplasms, fluorouracil in bile duct cancer, midazolam in metastatic disease) all sitting around 20.5%. These are the pipeline's "watch closely" programs: genuinely promising rates, but resting on so little evidence that the number could easily move a great deal with just one or two additional trials. They are not yet investable on the strength of this table alone, but they are exactly

the kind of candidates worth flagging for a standing watchlist.

The "*high risk high reward*" group produce the single most striking result in the whole table. The top program, latanoprost in glaucoma, posts a 20.7% rate on just four trials, on par with the best "*emerging signal*" entries, but every other program in this group belongs to a very different, much lower cluster: a set of six metabolic-disease programs (empagliflozin, vitamin D3, and insulin aspart in Phase 2, all tied at 12.6% on six trials each) followed by six more just behind them (atorvastatin, insulin glargine, resveratrol, insulin lispro, pramlintide acetate, and aspart again, each around 12.5% on five trials), spanning diabetes, obesity, and insulin resistance. The gap between latanoprost's 20.7% and this metabolic cluster's 12.5% is the widest jump anywhere in the table, and it is a useful reminder of what "*high risk high reward*" is actually built to capture: not necessarily a uniformly exciting group, but a profile that includes both a genuine standout (glaucoma) and a cluster of moderate, wide-uncertainty metabolic programs that simply haven't yet accumulated the evidence needed to settle into either the confident or the modest camp.

The two remaining groups, "*stable but modest*" and "*standard*", look almost identical to each other and, notably, both sit right around the 19.9-20% mark, close to the phase-wide Phase 1 average established in Section 5.2. That is exactly what their definitions predict: "*stable but modest*" programs are, by construction, reasonably well-evidenced (4 to 12 trials here, spanning a mix of oncology and non-oncology candidates such as ibuprofen for pain and laronidase for a rare metabolic disorder) but sitting at an unremarkable rate; "*standard*" programs, several of which rest on only a single supporting trial, are simply everything left over once the other four profiles have been carved out. Neither group is meant to surprise, their entire role in the table is to show what "ordinary" looks like, so that the standout numbers elsewhere are easier to recognize as standouts.

Read together, these five groups turn the single question "what are our best programs?" into a sharper, more useful one: "what does our best look like once we separate the sure things from the speculative ones?" The "*high confidence*" group answers where the safe capital should go. The "*emerging signal*" and "*high risk high reward*" groups answer where limited diligence time is best spent looking for the next confirmed winner, and the latanoprost result in particular is a concrete example of exactly the kind of thing this table is designed to surface: a single, genuinely striking candidate that a rate-only ranking would have buried beneath dozens of well-evidenced oncology programs.



Figure 2: Disease areas' EB progression rate for each of the four target phases.

Table 2: Top 10 programs by EB rate per strategic profile

RxCUI	MeSH ID	Drug name	MeSH descriptor	Phase	EB rate	CI lower	CI upper	Strategic profile	Trials
RX714438	D008268	pazopanib	macular degeneration	Phase1	0.2076	0.0996	0.3425	emerging signal	3
RX11256	D008269	vitamin e	macular edema	Phase1	0.2076	0.0996	0.3425	emerging signal	3
RX82122	D002386	levofloxacin	cataract	Phase1	0.2062	0.0985	0.3409	emerging signal	3
RX32592	D019337	oxaliplatin	hematologic neoplasms	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX70161	D008545	recombinant human gm-csf	melanoma	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX6960	D009362	midazolam	neoplasm metastasis	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX4492	D001650	fluorouracil	bile duct neoplasms	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX32592	D016410	oxaliplatin	lymphoma, t-cell, cutaneous	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX4451	D009837	flucytosine	oligodendroglioma	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX32592	D054198	oxaliplatin	precursor cell lymphoblastic leukemia-lymphoma	Phase1	0.2049	0.0976	0.3394	emerging signal	3
RX12574	D010190	gemcitabine	pancreatic neoplasms	Phase1	0.2709	0.1937	0.3557	high confidence	76
RX68446	D002289	pemetrexed	carcinoma, non-small-cell lung	Phase1	0.2536	0.1656	0.3532	high confidence	43
RX3264	D009101	dexamethasone	multiple myeloma	Phase1/2	0.2495	0.1725	0.3353	high confidence	72
RX342369	D009101	lenalidomide	multiple myeloma	Phase1	0.2477	0.1606	0.3467	high confidence	43
RX3002	D008545	cyclophosphamide	melanoma	Phase1	0.2476	0.1545	0.3544	high confidence	32
RX224905	D001943	trastuzumab	breast neoplasms	Phase1	0.2467	0.1532	0.3542	high confidence	31
RX3002	D015451	cyclophosphamide	leukemia, lymphocytic, chronic, b-cell	Phase1	0.2449	0.1504	0.3538	high confidence	29
RX337525	D002289	erlotinib	carcinoma, non-small-cell lung	Phase1	0.2430	0.1475	0.3534	high confidence	27
RX4492	D012004	fluorouracil	rectal neoplasms	Phase1	0.2430	0.1475	0.3534	high confidence	27
RX12574	D010190	gemcitabine	pancreatic neoplasms	Phase1/2	0.2426	0.1585	0.3381	high confidence	51
RX43611	D005901	latanoprost	glaucoma	Phase1/2	0.2072	0.0969	0.3460	high risk high reward	4
RX1545653	D009765	empagliflozin	obesity	Phase2	0.1257	0.0620	0.2077	high risk high reward	6
RX1244014	D014808	vitamin d3	vitamin d deficiency	Phase2	0.1257	0.0620	0.2077	high risk high reward	6
RX51428	D003920	aspart	diabetes mellitus	Phase2	0.1257	0.0620	0.2077	high risk high reward	6
RX83367	D003920	atorvastatin	diabetes mellitus	Phase2	0.1245	0.0608	0.2068	high risk high reward	5
RX274783	D003922	insulin glargine	diabetes mellitus, type 1	Phase2	0.1245	0.0608	0.2068	high risk high reward	5
RX1000492	D007333	resveratrol	insulin resistance	Phase2	0.1245	0.0608	0.2068	high risk high reward	5
RX86009	D003922	insulin lispro	diabetes mellitus, type 1	Phase2	0.1245	0.0608	0.2068	high risk high reward	5
RX356773	D009765	pramlintide acetate	obesity	Phase2	0.1245	0.0608	0.2068	high risk high reward	5
RX51428	D003922	aspart	diabetes mellitus, type 1	Phase2	0.1245	0.0608	0.2068	high risk high reward	5
RX4850	D009765	d-glucose	obesity	Phase1	0.1994	0.0947	0.3312	stable but modest	4
RX3002	D015266	cyclophosphamide	carcinoma, merkel cell	Phase1	0.1994	0.0947	0.3312	stable but modest	4
RX24698	D015266	fludarabine	carcinoma, merkel cell	Phase1	0.1994	0.0947	0.3312	stable but modest	4
RX6960	D018357	midazolam	respiratory syncytial virus infections	Phase1	0.1994	0.0947	0.3312	stable but modest	4
RX5640	D010146	ibuprofen	pain	Phase1	0.1992	0.0965	0.3277	stable but modest	6
RX6960	D009369	midazolam	neoplasms	Phase1	0.1992	0.1019	0.3193	stable but modest	12
RX4179	D006689	etoposide	hodgkin disease	Phase1	0.1992	0.0993	0.3233	stable but modest	9
RX1597582	D010190	olaparib	pancreatic neoplasms	Phase1	0.1992	0.0945	0.3309	stable but modest	4
RX392509	D008059	laronidase	mucopolysaccharidosis i	Phase1	0.1991	0.0945	0.3309	stable but modest	4
RX4179	D008527	etoposide	medulloblastoma	Phase1	0.1991	0.0945	0.3308	stable but modest	4
RX974779	D016403	obinutuzumab	lymphoma, large b-cell, diffuse	Phase1	0.1996	0.0938	0.3331	standard	3
RX2267582	D002295	trastuzumab deruxtecan	carcinoma, transitional cell	Phase1	0.1996	0.0916	0.3367	standard	1
RX284924	D054218	imatinib mesylate	precursor t-cell lymphoblastic leukemia-lymphoma	Phase1	0.1996	0.0916	0.3367	standard	1
RX2267582	D009362	trastuzumab deruxtecan	neoplasm metastasis	Phase1	0.1996	0.0916	0.3367	standard	1
RX194337	D002292	vorinostat	carcinoma, renal cell	Phase1	0.1996	0.0938	0.3331	standard	3
RX25102	D007947	fludarabine phosphate	leukemia, megakaryoblastic, acute	Phase1	0.1996	0.0938	0.3331	standard	3
RX25102	D004915	fludarabine phosphate	leukemia, erythroblastic, acute	Phase1	0.1996	0.0938	0.3331	standard	3
RX1232150	D057135	aflibercept	wet macular degeneration	Phase1	0.1996	0.0916	0.3367	standard	1
RX3264	D001752	dexamethasone	blast crisis	Phase1	0.1996	0.0916	0.3367	standard	1
RX3002	D009373	cyclophosphamide	neoplasms, germ cell and embryonal	Phase1	0.1996	0.0916	0.3367	standard	1

6 CONCLUSION

This project set out to answer a central question in clinical development: given a drug candidate at a particular phase, for a given disease, how likely is it to advance, and how confident should we be in that estimate? Starting from nothing more than public NLM CTGov registrations, the NLM MeSH ontology, and a NLM RxNorm vocabulary subsample, the pipeline resolves messy drug and disease identities, models progression with a survival framework suited to censored, low-event-rate data, shrinks noisy small-sample rates via Empirical Bayes, and turns the result into a ranked, profiled table of 51,388 real drug-disease-phase programs.

The results behave the way a domain-literate reviewer would expect: progression probability declines with phase, oncology dominates by volume, and the strongest well-evidenced candidates are mature

compounds rather than statistical noise, while the pipeline’s risk-stratified view also surfaces genuine standouts, like latanoprost’s high rate on thin evidence, that a single rate-only ranking would have hidden. Just as importantly, this paper has been explicit that these outputs measure a narrower signal than a published probability-of-success estimate, and the observed gap (lower rates than literature benchmarks) runs in exactly the direction the stated limitations predict.

One closing observation is worth making explicit: the top programs table (Section 5.3) is organized precisely so that the safest bets, “*high confidence*” programs, high rate and low uncertainty alike, do not crowd out everything else. The more commercially interesting segments often sit in the other four groups. The clearest example produced by this run is latanoprost in glaucoma: a “*high risk high*

reward" program posting a 20.7% progression rate on just four supporting trials, on par with the pipeline's strongest early-phase oncology candidates but resting on a fraction of the evidence. That is exactly the kind of candidate where a modest amount of additional diligence, one or two more studies, could resolve real uncertainty on an already-promising rate, making it a natural in-licensing or co-development target before the rest of the market catches up. The *"emerging signal"* group, meanwhile, holds several programs sitting at 20-21% on only three supporting trials each, pazopanib in macular degeneration and vitamin E in macular edema chief among them, too thin on data to act on today, but worth flagging for a standing watchlist that re-scores automatically as new trials accumulate. Both groups sit outside the *"high confidence"* core precisely because they are less certain, but for a strategy team whose mandate is finding under-the-radar opportunity rather than confirming the obvious, it is these subgroups, not the well-evidenced oncology names at the top, where this pipeline's output is likely to add the most differentiated value.

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